



FROM ZERO TO PRO - MODULE 09

Baking & Pastry Fundamentals

Baking is chemistry with deadlines; precision, ratios, and a scale replace improvisation.

Weeks 15-16 - watch, study, cook, reflect

By the end of this module

- 1 Work by weight; understand baker's percentages.
- 2 Make a basic bread dough and understand gluten and fermentation.
- 3 Produce pie dough, a simple cake, and a custard dessert.
- 4 Understand leavening: yeast, chemical, and mechanical.

LESSON 9.1

Precision and Baker's Percentages

Objective: Work by weight and understand baker's percentages.

Key points

- Weigh ingredients in grams; volume measures are unreliable for baking.
- Flour = 100 percent; every other ingredient is a percent of flour weight.
- Hydration = water as a percent of flour; it controls dough texture.
- A scale makes recipes repeatable and easy to scale.

Watch out for

- Measuring flour by the cup and getting wildly different amounts.
- Eyeballing leaveners and salt, which need precision.
- Ignoring ingredient temperature, which affects doughs and batters.

LESSON 9.2

Bread and Gluten

Objective: Make a basic dough and understand gluten and fermentation.

Key points

- Gluten is the stretchy protein network that gives bread structure.
- Develop it by kneading or by fold-and-rest cycles.
- Fermentation builds flavor; slower and cooler is deeper.
- Proof until puffy and jiggy, then bake hot to set crust and crumb.

Technique

- 1 Mix**
Combine flour, water, salt, and yeast into a shaggy dough.
- 2 Develop**
Knead or do stretch-and-folds to build gluten.
- 3 Ferment**
Let it rise until roughly doubled and airy.
- 4 Shape and bake**
Shape, proof, and bake in a hot oven.

LESSON 9.3

Pie and Tart Dough

Objective: Make a flaky, all-butter dough and blind-bake a shell.

Key points

- Keep the butter cold and in pieces for flaky layers.
- 3-2-1 by weight: flour, fat, water.
- Add minimal water and do not overwork the dough.
- Rest cold before rolling; blind-bake shells for wet fillings.

Technique

- 1 Cut in fat**
Cut cold butter into flour until pea-sized pieces remain.
- 2 Add water**
Sprinkle in just enough ice water to bring it together.
- 3 Rest**
Chill the disc to relax gluten and firm the butter.
- 4 Roll and chill**
Roll evenly, shape, chill, then blind-bake if needed.

LESSON 9.4

Cakes and Custards

Objective: Bake a tender cake and set a simple custard dessert.

Key points

- Creaming whips air into butter and sugar for a tender crumb.
- Do not overmix after adding flour, or the cake turns tough.
- Bake custards gently, often in a water bath, for even setting.
- Custard is done when set at the edges but still jiggly in the center.

Watch out for

- Overmixing batter and getting a dense, rubbery cake.
- Baking custard too hot, so it curdles and weeps.
- Opening the oven early and collapsing a delicate cake.

Practice & homework

1

Bake a simple loaf by weight; log hydration and crumb.

2

Make an all-butter pie dough and blind-bake a shell.

3

Bake one cake by the creaming method and evaluate crumb.

Watch & learn (free)



Brian Lagerstrom (ex-restaurant pro, approachable)

<https://www.youtube.com/@BrianLagerstrom>



America's Test Kitchen (tested technique)

<https://www.youtube.com/@AmericasTestKitchen>



ChefSteps (modern technique & sous vide)

<https://www.youtube.com/channel/UCxD2E-bVoUbaVFL0Q3PvJTg>



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and a scale replace improvisation.***

Next up - Module 10: Global Cuisines: French, Italian, Asian & Latin

FROM ZERO TO PRO - THE SELF-TAUGHT CHEF PROGRAM